Sam Meydanshahi

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Education

University of Waterloo Waterloo, ON
Bachelor of Computer Science (Honours) *2021 – 2026*
Coursework: Algorithms, Operating Systems, Machine Learning, Object-Oriented Programming, Functional Programming, Distributed Systems, Databases, Concurrency, Networks, Data Structures, Compilers

Technical Skills

Languages: C++, Python, Java, Go, Rust, JavaScript, TypeScript, Kotlin, Swift, SQL
Technologies & Tools: React, Angular, Next.js, Node.js, Spring Boot, Django, FastAPI, PyTorch, TensorFlow, LangChain, Docker, Kubernetes, AWS, GCP, PostgreSQL, MongoDB, Supabase, pgvector, Redis, CI/CD, RAG

Experience

Tenstorrent January 2026 – August 2026
Software Engineer Intern *Toronto, Ontario, CA*
- Optimized the tensor tilize operator by refactoring the data transformation pipeline, achieving a **77% reduction in operation latency** and directly improving LLM inference throughput
- Diagnosed & fixed L1 data corruption across **9 dataflow kernels** caused by incorrect circular-buffer pointer usage
- Optimized the reshape operator by eliminating intermediate sharded-to-interleaved and interleaved-to-sharded conversions for tiled inputs, **reducing operation latency by 21%**
- Authored 2D-fabric CCL tests and end-to-end MoE inference tests on a 32-chip Galaxy system, integrating them into post-commit & nightly CI to **catch distributed multi-chip LLM inference regressions early**
- Built a memory sanitizer for AI accelerator kernels, detecting silent hardware bugs (OOB writes, buffer misuse, synchronization violations) in emulation; built dedicated testing in CI, finding bugs in **20+ production kernels**
- Unified the cross-device back-prop handshake used by all LLM inference pipeline stages into a shared abstraction, and overlapped incoming data transfer with the handshake wait to **hide inter-chip synchronization latency**

Payments Canada May 2025 – August 2025
Software Engineer Intern *Ottawa, Ontario, CA*
- Developed and optimized participant search functionality (Angular + Spring Boot) in the RTR payments system, improving query flexibility and reducing errors for **100+ financial institutions**
- Redesigned REST API response payloads across RTR and FIF systems to return richer, consolidated data, reducing frontend API calls and **improving load performance by 25%**
- Implemented robust data validation and error-handling middleware in RTR and FIF APIs, **resolving 4 recurring 500-level errors** and preventing duplicate submissions or intentionally manipulated user inputs

Teranet January 2024 – April 2024
Software Engineer Intern *Toronto, Ontario, CA*
- Implemented core components of GeoWarehouse, **optimizing load times for 100,000+** real estate professionals
- Refactored and optimized database search query logic through indexing and query restructuring, **improving query efficiency by 25%**

Teranet January 2023 – April 2023
Software Engineer Intern *Toronto, Ontario, CA*
- Built a full-stack internal dashboard using Angular, TypeScript, and Node.js to **monitor 30+ backend services**, improving DevOps visibility and **cutting manual status checks by 75%** and improving DevOps incident response

Technical Projects

Aware In Medicine | *Next,js, TypeScript, Supabase, pgvector, OpenAI Embeddings, Anthropic API, PostgreSQL*
- Built and deployed a nonprofit medical education platform with a conditions library, weekly content system, and an AI semantic search via a RAG pipeline over a pgvector store (OpenAI embeddings, Claude API)

AI Real Estate Assistant | *Node.js, Express, Claude API, Vanilla JS, Render*
- Built an AI assistant for a real estate client for natural language property Q&A, deployed as an embeddable widget